Koneru Lakshmaiah Education Foundation
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I/IV B. Tech. Even Semester: A.Y.2025-26

Linear Algebra & Calculus for Engineers (25MT1001)

CO-2 TUTORIAL PROBLEMS

TUTORIAL-4: Partial derivatives, Jacobian, total derivative, chain rule, Taylor's series for two variables

1. Compute the first and second order partial derivatives of $z = x^2y^3 + \sin x \cos y$.
2. Given $u = \log(x+y+z)$, $x = e^t, y = \sin t, z = \cos t$ then find the total derivative $\frac{du}{dt}$
3. If $z = xy$ where $x = s^2 + t^2$, $y = 2st$, Evaluate $\frac{\partial z}{\partial s}$, $\frac{\partial z}{\partial t}$ by using chain rule.
4. If $u = x^2 - 2y, v = x + 2y + z^2, w = x - 2y^2 + 3z$ find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at (1,0,0).
5. Apply Taylor's series to expand $f(x,y) = x^3 + 2xy + y^3$ in powers of $(x+1)$ and $(y+2)$ up-to second degree terms.
6. Expand the function f (x, y) =e^x log(1+y) in terms of 'x' and 'y' up to second degree terms using Maclaurin's series.

TUTORIAL-5: Maxima and minima, Lagrange's method of multipliers

1. Determine the maxima and minima of $f(x,y) = x^2 + y^2 + 8x + 4$.
2. Illustrate extremum values of $f(x,y) = x^2 + y^2 + 2x + 4y + 1$.
3. Evaluate extrema of $f(x,y) = x^3 + y^3 - 3xy$.
4. The sum of three numbers is 6. Prove that their product is maximum when they are equal.
5. A rectangular box open at the top is to have a volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction
6. The temperature T at any point (x, y, z) in space is given by $T = xyz^2$. Find, the highest temperature on the surface of the unit sphere of $x^2 + y^2 + z^2 = 1$.

TUTORIAL-6: Solving second and higher order differential equations and its applications

1. Solve the ODE $\frac{d^3y}{dx^3} - 9\frac{d^2y}{dx^2} + 23\frac{dy}{dx} - 15y = 0.$
2. Solve the O DE $y''' + 4y'' + 4y' = 0$.
3. Determine the solution of $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 5$, given that y(0) = 0, $y'(0)=1$
4. Solve the differential equation $(D^2 - 4)y = \cosh 5x$
5. The motion of a mass spring system with damping and external force is described by $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = e^{-4t}, \boldsymbol{x(0) = 0, x'(0) = 1}$, where x is the distance of the mass from the equilibrium position, downward being taken as positive direction. Calculate the displacement '$x(t)$' of the motion.
6. The motion of a mass spring system without damping and external force is described by $\frac{d^2x}{dt^2} + 4x = Sin\, 3t, \boldsymbol{x(0) = 0, x'(0) = 0}$, where x is the distance of the mass from the equilibrium position, downward being taken as positive direction. Evaluate displacement '$x(t)$' of the motion.
7. Determine the charge **'q(t)'** on the capacitor in an **LRC** series circuit at t when inductance 1 H, resistance 4**Ω**, capacitance 0.25 F, E(t) = 0 V, q(0) =5 C, and i(0) =0 A.
8. Determine the **charge 'q(t)' and current 'i(t)'** on the capacitor in an LC series circuit at t when inductance 1 H, capacitance 1F, E(t) = e3t V, q(0) = 2 C,and i(0) = 0 A.